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EDITORIAL

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Next stop, precision: targeted therapies for *Staphylococcus aureus* infections

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1. Precision therapies: a missing link in infectious disease

In recent decades, precision medicine has transformed oncology by tailoring treatments to individual genetic profiles and tumor phenotypes [1]. In contrast, infectious disease management continues to rely heavily on broad-spectrum antibiotics, often administered empirically without confirmed pathogen identification [2]. This generalized approach contributes to the growing crisis of antimicrobial resistance while also disrupting the host microbiome [3].

S. aureus exemplifies the challenges of this outdated model. As a pathogen responsible for a wide spectrum of illnesses, ranging from minor skin infections to life-threatening sepsis, it has evolved multiple mechanisms to evade both host immunity and antibiotic action [4]. The rise of methicillin-resistant S. aureus (MRSA) and vancomycin-resistant S. aureus (VRSA) underscores the urgent need for targeted approaches [5,6]. Traditionally, antimicrobial development has focused on discovering new antibiotic agents, while delivery and pharmacokinetics have been considered secondary concerns [7]. However, an emerging "delivery first" paradigm suggests that developing effective and specific delivery platforms can significantly enhance the impact of both existing and novel antimicrobial agent [8]. This approach shifts the focus from improving drug potency to ensuring that drugs reach the infection site in a targeted and efficient manner. By exploiting pathogen-associated features or infected tissue environments, this strategy enhances therapeutic precision, increases local drug concentration, reduces systemic toxicity, and preserves the microbiome. Ultimately, "delivery first" enables flexible and modular development, allowing validated delivery platforms to be paired with a range of drug payloads for specific clinical needs [9].

2. *S. aureus* as a model for precision antimicrobials

Due to its clinical significance and complex resistance mechanisms, *S. aureus* serves as a strong model for developing precision-based antimicrobial platforms [10]. Its ability to form biofilms and acquire drug resistance presents major challenges to conventional therapies [4]. Three promising approaches include bacteriophage therapy, ADCs and TOUCANs, each of which offers a distinct strategy for pathogen-specific antimicrobial delivery, will be discussed in the following sections.

2.1. Bacteriophage therapy: tailored but complex

Bacteriophages, or phages, are viruses that infect and lyse bacteria with high specificity [9]. They were widely used before antibiotics became standard care [9]. As resistance to traditional drugs increases, phage therapy is receiving renewed attention [11]. Unlike broad-spectrum antibiotics, phages typically infect only a narrow range of bacterial strains, enabling selective eradication of pathogens like *S. aureus* without disrupting commensal microbiota. Phage cocktails, combining multiple phages with complementary host ranges, increase efficacy and reduce resistance risk [12]. Moreover, phage-encoded endolysins, which degrade bacterial cell walls, have shown strong potential against *S. aureus* [13]. Phage therapy holds promise for difficult cases such as biofilm-associated or multidrug-resistant *S. aureus* infections [13,14]. Advances in synthetic biology have expanded the potential of phages. Engineered phages can now be optimized for stronger lytic activity, broader host range, or even designed to deliver enzymes or gene-editing tools [11].

Despite its potential, phage therapy still faces several challenges, including strain-specific matching that complicates standardization, immune neutralization during systemic infections, inconsistent clinical outcomes, and difficulties with large-scale manufacturing [11]. Regulatory frameworks for phage therapeutics are also evolving slowly. While early clinical studies are encouraging, phage therapy is not yet widely implemented [14]. Nevertheless, phages provide a compelling model of strain-level specificity that aligns with the goals of precision antimicrobial development.

2.2. Antibody–drug conjugates (ADCs): precision from oncology to infection

ADCs are well-established in oncology, where monoclonal antibodies are used to deliver cytotoxic agents directly to

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cancer cells [15]. This strategy is now being adapted to infectious diseases, with efforts underway to target pathogens such as *S. aureus* [16]. In this context, ADCs use antibodies that recognize conserved bacterial surface components, such as wall teichoic acids or protein A. Once bound, the drug payload is either internalized or released in close proximity to the pathogen, improving local efficacy while reducing off-target toxicity. This approach is particularly useful for infections involving biofilms or bacteria reservoirs, where free antibiotics often fail to reach effective concentrations [16].

Several preclinical studies have shown that *S. aureus*-targeted ADCs can reduce bacterial burden in complex models, including prosthetic joint infections [17] and endocarditis [18]. These conjugates improve tissue penetration, increase drug concentration at the infection site, and minimize systemic exposure [18].

However, significant challenges remain. These include identifying bacterial antigens that are both specific and conserved across strains, engineering linkers that release the drug effectively in the infection microenvironment, and minimizing the immunogenicity of the antibody itself. Manufacturing complexity and high costs may also limit short-term scalability [17,18]. Despite these issues, ADCs represent a promising conceptual advance, adapting principles from oncology to the field of infectious diseases.

2.3. TOUCAN: a nanocarrier delivery system approaching clinical readiness

Among the emerging precision platforms, the TOUCAN system (Therapeutic OligonUCleotides Activated by Nucleases) stands out for its clinical feasibility and modular design. Its first candidate, TOUCAN-MRSA, targets *S. aureus*, including methicillin-resistant strains, using a delivery-first approach to achieve highly specific antimicrobial action [19]. TOUCAN uses short synthetic oligonucleotides (7 to 10 nucleosides in length) as a structural scaffold. These are conjugated to therapeutic agents such as nucleoside analogues, small-molecule antibiotics, or immune-modulatory drugs. The nanocarrier is designed to exploit bacterial virulence factors, especially associated nucleases [20], as both recognition motifs and enzymatic triggers for payload release [19].

TOUCAN-MRSA remain extracellular due to their phosphodiester backbone but release their payload upon encountering specific *S. aureus* nucleases in the infection microenvironment. The released payload is then selectively internalized via bacterial transport pathways, enabling localized action while sparing host tissues and microbiota [19]. TOUCAN addresses several critical therapeutic barriers. It enables targeted delivery of drug to the infection microenvironment, avoids the need for high plasma concentrations, and concentrates therapeutic activity where it is most needed. The modular design allows for payload customization without altering the core delivery structure. Unlike biologics, TOUCAN's chemically defined composition facilitates scalable and reproducible pharmaceutical manufacturing within existing regulatory frameworks [19].

Importantly, this platform reflects a broader shift in strategy [21]. By prioritizing delivery first [8], technologies such as TOUCAN separates the development of uptake and targeting from payload optimization. Once pathogen-specific delivery is validated, therapeutic innovation becomes more flexible and efficient. TOUCAN-MRSA represents a tangible step toward clinical implementation of this type of concept.

3. Outlook: what comes after delivery?

Precision strategies are no longer an academic exercise or a technological trend. They are a necessary response to the escalating threat of antimicrobial resistance. The time for patching old antibiotics is over. We must now invest in approaches that extend the clinical utility of new drugs and minimize the need for broad-spectrum agents.

Targeted delivery can fundamentally change how we treat infections. Rather than relying on ever-wider antibiotic spectra or new chemical scaffolds, researchers can focus on delivering drugs to the right place at the right time. Once a delivery platform is in place, a range of payloads becomes viable, including traditional antibiotics, metabolic inhibitors, and immune modulators. Precision will not replace antibiotics, but it will dictate how and where they are used.

Antimicrobial resistance is a multifactorial challenge, and pathogens like *S. aureus* employ diverse survival tactics, including drug efflux, immune evasion, biofilm formation, etc. No single solution will overcome all of these obstacles. However, precision delivery strategies serve as powerful enablers, increasing the specificity and potency of the tools we already have.

The "delivery first" mind-set also provides practical advantages. It reduces risk in the development pipeline by validating selective uptake mechanisms before optimizing the therapeutic compound. This is particularly important for diseases with challenging pharmacokinetics, such as intracellular or biofilm-associated infections.

In the future, precision antimicrobials may combine various approaches. Nanocarriers, phages, ADCs, and gene-targeting tools may be integrated into multi-faceted therapies. Successful clinical adoption will require advances in rapid diagnostics, manufacturing, and regulatory infrastructure. Yet the underlying principle remains the same. The next chapter in infectious disease treatment will begin with how and where we deliver the therapy.

4. Conclusion: precision is the future

S. aureus remains one of the most persistent and difficult pathogens in modern medicine. Its ability to adapt, resist, and evade continues to challenge conventional antimicrobial strategies. Precision therapies offer a new direction, not by replacing antibiotics, but by reimagining their delivery and specificity.

As outlined in this editorial, three promising approaches, including bacteriophage therapy, ADCs and TOUCANs, illustrate the growing momentum toward targeted intervention for infectious diseases. Embracing delivery-first strategies, supported by specific targeting and modular design, will help close the gap between innovation and implementation. The future of antimicrobial therapy will not be uniform; it will be intelligent, selective, and strategically delivered. *S. aureus*

could be the first pathogen to demonstrate just how far this new model can go.

Disclosure statement

The author is the inventor of a patent covering the TOUCAN approach and is a co-founder of a spin-off company that uses TOUCAN as its core technology.

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